Mohamed Shamseldein

PhD, P.Eng. | Senior Member, IEEE

Power System Engineering Leader | Data Centres and Large Loads

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Professional profile

Power-system engineering leader with more than 13 years at the Independent Electricity System Operator (IESO) and earlier research and teaching experience dating to 2003. Leads technical work on large computational loads, connecting facility behaviour to grid reliability and connection requirements. Combines Ontario system-operator experience with industry collaboration, renewable-energy research and mentoring. Registered Professional Engineer in Ontario.

Selected qualifications

- Led development of IESO technical requirements for large computational loads, addressing disturbance ride-through, restoration, demand variability, ramping, oscillations, monitoring and modelling.
- Led the power-system impacts team within ESIG's large-load initiative and served as a lead author of *Reliability Impacts of Large Loads* (2026).
- Received Energy Storage Canada's 2026 Foundational Research in Energy Storage Award for research on battery-supported mitigation of data-centre grid reliability risks.

Professional experience

Independent Electricity System Operator

Principal Power System Engineer, Large Loads Integration | 2026 to present

- Lead technical work on large computational and other emerging loads, translating performance and modelling needs into requirements that support connection assessment.
- Coordinate engineering perspectives across connection stakeholders and mentor engineers on facility behaviour, model assumptions, study interpretation and reliability evidence.

Progressive power-system engineering roles | August 2013 to 2026

- Assessed and reviewed connections involving generation, battery storage, transmission and large loads, including inverter-based resources.
- Conducted and guided reliability studies and technical reviews using PSS/E, PSCAD and PowerTech DSA; worked on system operating limits and assessment evidence.
- Contributed to the Lakeshore Transmission Station Integration project and the Ontario Resource and Transmission Assessment Criteria update, with IESO team recognition.

Industry leadership and knowledge sharing

ESIG Large Loads Task Force. Power-system impacts team lead and a lead author of the 2026 reliability-impacts report. Contributed engineering knowledge through collaboration across utilities and industry organizations.

NERC. IESO member of the Project 2023-05 drafting team for modifications to FAC-001 and FAC-002. Acknowledged contributor to the March 2026 white paper on gaps in practices, requirements and reliability standards for emerging large loads.

Technical communication. Presented *Integration of Large Loads to the Ontario Power System* at the 2026 CIGRE Energy Forum in Calgary. Invited speaker and panel participant at the 2025 IEEE SmartGridComm workshop *AI for Energy & Energy for AI*.

Selected research relevant to the role

- **Data-centre storage and controls.** Sole-authored research on supervisory control of centralized UPS systems for grid-demand management and fault response, evaluated through simulation. *IET Generation, Transmission & Distribution*, 2026. doi:10.1049/gtd2.70369.
- **Integrated computing and energy loads.** Research on the connection assessment of co-located AI data centres, hydrogen production and battery storage. *Global Energy Interconnection*, 2026. doi:10.1016/j.gloei.2026.08.002.
- **Renewable generation.** Research on photovoltaic-array optimization and wind-generator fault response, including publications in *IEEE Transactions on Sustainable Energy* and *Electric Power Systems Research*.

Recognition

Energy Storage Canada Foundational Research in Energy Storage Award, 2026. ESIG Excellence Award, 2026, as one of five engineers recognized for collaborative large-load integration and knowledge sharing. IESO Team Beacon recognition for Lakeshore integration; Beacon recognition for contributions to the ORTAC update during 2025.

Education and leadership development

PhD, Electrical and Computer Engineering | University of Waterloo | 2012
MSc, Electrical Power Engineering | Ain Shams University | 2007
BSc, Electrical Power Engineering | Ain Shams University | 2003

Harvard Business School Online: Leadership Principles; Management Essentials; Power and Influence for Positive Impact (2023-2024). Ivey Business School Executive Education / IESO New People Leaders Program (credential issued May 2024).

Academic experience and professional standing

Assistant Professor, Ain Shams University, 2016-present; on approved leave, with IESO as primary employer. University of Waterloo lecturer and research/teaching assistant, 2008-2013; Ain Shams University research/teaching assistant, 2003-2008.

Professional Engineer, Ontario | Senior Member, IEEE | IEEE Power & Energy Society
References available upon request.